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25 August 1981
SUPERSEDING
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27 October 1979

MILITARY SPECIFICATION

CLOTH, TAFFETA, NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This specification covers the requirements for nylon taffeta cloth suitable both as a base fabric without coating or for coating with chloroprene rubber or other type coatings. (See 6.5).

1.2 Classification The cloth shall be of the following types as specified (see 6.2):

Type I	-	1.2 oz/yd^2 ($41g/m^2$)
Type II	-	1.5 oz/yd^2 ($51g/m^2$)
Type III	-	2.0 oz/yd^2 ($68g/m^2$)
Type IV	-	1.9 oz/yd^2 ($64g/m^2$)

2. APPLICABLE DOCUMENTS

2.1 Issue of documents The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-295	- Thread, Nylon
PPP-P-1133	- Packaging and Packing of Synthetic Fiber Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing & Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

Rules and Regulations under the
Textile Fiber Product Identification Act

(Application for copies should be addressed to the Federal Trade Commission, Washington, D.C. 20580.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issues in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials
D1424 Test for Tear Resistant of Woven Fabrics by
Falling-Pendulum (Elmendox) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Standard sample The cloth shall match the standard sample for shade, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article When specified, the contractor shall furnish a sample of cloth for first article inspection and approval (see 4.3 and 6.2).

3.3 Material

3.3.1 Fiber

3.3.1.1 Type I, II and IV The fiber shall be a polyamide prepared from hexamethylene diamine and adipic acid or its derivatives and shall have a minimum melting point of 472° F (244° C) when tested as specified in 4.4.1.

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3.3.1.2 Type III The fiber shall be first quality semi-dull regular tenacity nylon when tested as specified in 4.4.1.

3.3.2 Yarn

3.3.2.1 Type I The yarn shall be multifilament, nominal 50 denier (5.6 Tex) for the warp and filling, when tested as specified in 4.4.1.

3.3.2.2 Type II The yarn shall be multifilament nominal 50 denier (5.6 Tex) for the warp and nominal 70 denier (7.6 Tex) for the filling when tested as specified in 4.4.1.

3.3.2.3 Type III and Type IV The yarn shall be continuous filament, nominal 70 denier (7.6 Tex) for both the warp and filling, when tested as specified in 4.4.1.

* 3.4 Physical requirements The finished cloth shall conform to the requirements in Table I when tested as specified in 4.5.

Table I - Physical Requirements

Characteristic	Type I	Type II	Type III	Type IV
Weight oz/ sq yd (g/m^2)	1.2 (41) (max)	1.5 (51) (max)	Min 2.0 (68) Max 2.3 (78)	1.9 (± 0.15) (64 ± 5)
Yarns per inch (min) (per cm)				
Warp	100 (39.4)	95 (37.4)	106 (41.7)	108 (43.2)
Filling	66 (26.0)	65 (25.6)	92 (36.2)	71 (28.4)
Breaking Strength lbs (Newtons) min.				
Warp	80 (356)	75 (334)	110 (489)	110 (489)
Filling	55 (245)	75 (334)	95 (423)	85 (378)
Tearing Strength, lbs (Newtons) min				
Warp	4.0 (18)	4.0 (18)	3.64 (16)	5.0 (22)
Filling	4.0 (18)	6.0 (27)	2.97 (13)	3.5 (16)
Air permeability, CFM/sq ft (cm^2 / sec/cm^2)	120 ± 20 (61.0 ±10.2)	70 ± 15 (35.6 ±7.6)	-	-
Flex stiffness, in-lbs X 10 $^{-4}$max (N.m) 1/				
Warp			1.5 (0.17)	
Filling			0.5 (0.06)	

1/ When specified (see 6.3)

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3.4.1 Weave The weave shall be a taffeta (plain weave).

3.4.2 Width The width of the cloth shall be as specified (see 6.2) and shall be the minimum width, inclusive of selvage. When cloth has shuttleless loom selvages, the width of the cloth shall be as specified (see 6.2) and shall be minimum width of the woven cloth, excluding the selvages.

3.5 Color The color of the dyed cloth shall be as specified (see 6.2) and shall match the standard sample.

3.5.1 Matching The color of the finished cloth shall match the standard sample under artificial daylight, having a color temperature of 7500° ($\pm$ 100) Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° ($\pm$ 100) Kelvin.

3.5.2 Colorfastness

3.5.2.1 Type I and II The finished cloth shall show fastness to light, water and crocking equal to or better than the standard sample. When no standard sample is available the cloth shall show a minimum of "good" fastness to light and water and a Munsell value for crocking not lower than 8.5 when tested as specified in 4.5.

3.5.2.2 Type III and IV The finished cloth shall show fastness to light, laundering, perspiration, dry cleaning and crocking equal to or better than the standard sample. When no standard sample is available, the cloth shall show a minimum of "good" fastness to light, laundering, perspiration and dry cleaning and a Munsell value for crocking not lower than 8.5, when tested as specified in 4.5.

3.6 Length and put up Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards (36.4m) for type III and IV cloth and 75 yards (68.2m) for Type I and II cloth. Each length shall be put up on a full width rolls as specified in PPP-P-1133.

3.7 Finish The cloth shall be scoured and heat set. The Types I, II and IV cloths may be calendered. Type III cloth shall not be calendered.

3.7.1 Permanence of finish (type I and II) The cloth finish shall have the following requirements when tested as specified in 4.5:

a. The air permeability readings taken after testing shall be within 15 percent of the readings taken before testing.
b. The cloth thickness after testing shall not exceed 10 percent more than before testing.
c. The shrinkage after testing shall not exceed 2 percent in either the warp or filling direction.

3.7.2 Heat treating (Types III and IV) The cloth shall be fully heat set at a minimum temperature of 400°F (204°C). The cloth shall show no appreciable change in color, distoration, or puckering and not more than 2.0 percent dimensional change in either the warp or the filling when tested as specified in 4.5.

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3.7.3 Nonfibrous materials The finished cloth shall not exceed 1.0 percent (Types III and IV) and 2.0 percent (Types I and II) starch and protein content including chloroform-soluble and water-soluble material when tested as specified in 4.5.

3.7.4 pH The pH of the cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.8 Seam slippage (Types I and II) The load required to separate the seam 1/2 inch (1.3cm) shall be as stated below in either the warp or filling direction when tested as specified in 4.5:

Type I - 10 pounds (44N) (min)
Type II - 20 pounds (89N)(min)

3.9 Fiber identification Each roll of cloth shall be labeled, or ticketed for fiber content in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.9.1 Marking The face side of the cloth shall be identified by stamping on that side with the word "FACE" at each end of each piece.

3.10 Workmanship The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards ($83.6m^2$), when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection When required, the first article of the finished cloth submitted in accordance with 3.2 shall be visually inspected for appearance and color. The sample shall be tested for physical and chemical properties in accordance with the methods specified in 4.5.

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4.4 Quality conformance inspection Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document. In addition, the supplier shall furnish a certificate of compliance for the nylon fiber and yarn requirements of 3.3.1 and 3.3.2 and the heat treating temperature requirement of 3.7.2.

4.4.2 Examination of the end item Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.5.

4.4.2.1 Yard-by-yard examination Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yard (91.4m) the entire yardage on the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4m), only 100 yards shall be examined. Defects as defined in Section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) (.9m) shall be scored and assigned demerit points as listed in 4.3.2.2 except that only those knots which exceed the limit shown on Sears Fabric Defect Scale (see 6.4), "B" for knots, shall be scored. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of the total yardage examined exceeds the following:

30 points for Types I, II and IV cloth
16 points for Type III cloth

The lot shall be unacceptable if the points per 100 square yards ($83.6m^2$) of two or more individual rolls exceeds the following:

45 points for Types I, II and IV cloth
24 points for Type III cloth

If one roll exceeds the point level, a second sample of the same number of rolls as in the first sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards } (83.6m^2)$$

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4.4.2.2 Demerit points Demerit points shall be assigned as follows:

For defects up to and including 3 inches (7.6cm) in any dimension	- one point
For defect exceeding 3 inches (7.6cm) but not exceeding 6 inches (15.2cm) in any dimension	- two points
For defects exceeding 6 inches (15.2cm), but not exceeding 9 inches (22.9cm) in any dimension	- three points
For defects exceeding 9 inches (22.9cm) in any dimension	- four points

The following defects when present, shall be scored 4 points for each yard (.9m) in which they occur:

Baggy, ridgy or wavy cloth
Poor dye penetration, mottled, streaky, or cloudy
Width less than specified
Holes, cuts, tears, open places

4.4.2.3 Examination for length

4.4.2.3.1 Individual rolls During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards (1.8m) less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.2.3.2 Total yardage in sample The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

* 4.4.2.4 Examination for shade variation During the yard-by-yard examination, each roll in the sample shall be examined for shade variation. Any roll in the sample exhibiting uneven shade, shade variation side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

4.4.2.5 Examination for face identification and non-conformance with the Textile Fiber Products Identification Act During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends or not labeled or ticketed in accordanc with the Textile Fiber Products Identification Act.

4.4.3 Examination of packaging requirements An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking comply with Section 5 requirements of this specification.

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4.5 Testing of the end item The methods of testing specified in FED-STD-191 wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test methods. All test reports shall contain the individual values utilized in expressing the final result. The sample unit for the test purposes shall be 2 continuous yards (1.8m) for Type I, II and IV cloth and 3 1/2 continuous yards (3.2m) for Type III cloth full width of the finished cloth. The lot size shall be expressed in units of one yard (.9m). The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test Methods

Characteristic	Requirement Paragraph	Test Method
Weight	3.4	5041
Yarns per inch (cm)		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength		
Warp	3.4	5100
Filling	3.4	5100
Tear Strength		
Warp	3.4	ASTM-D1424-63 1/
Filling	3.4	ASTM-D1424-63 1/
Flex Stiffness		
Warp	3.4	5206
Filling	3.4	5206
Air permeability	3.4	5450
Weave	3.4.1	Visual 2/
Colorfastness to:		
Laundering	3.5.2	5614 3/
Crocking	3.5.2	5651
Perspiration	3.5.2	5680
Dry Cleaning	3.5.2	5620
Light	3.5.2	5660 4/
Water	3.5.2	5630
Permanence of finish		
Air permeability	3.7.1	4.5.2
Thickness	3.7.1	4.5.2
Shrinkage	3.7.1	4.5.2
Heat Treating	3.7.2	4.5.1
Nonfibrous Materials	3.7.3	2611
pH	3.7.4	2811
Seam slippage	3.8	5420 5/

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1/ Standard instrument with one augmenting weight.
2/ One determination per sample unit and the result reported as "pass" or "fail".
3/ Only the stain on the nylon fibers of the color transfer cloth shall be evaluated.
4/ Total exposure time shall be 20 hours for Type III cloth only.
5/ The following exception to method 5420 shall be made:

a. Test machine shall have a maximum capacity of 75 pounds (34.1Kg)
b. The use of 2 inch (5.08cm) front jaws is permissible
c. Type I, Class 1, size E nylon thread conforming to V-T-295
d. Eight stitches per inch shall be used

4.5.1 Heat resistance The test specimen shall consist of a square of cloth at least 12 by 12 inches (30.5 by 30.5cm). It shall be laid flat without tension, and marked off with a square 10 by 10 inches (25.4 by 25.4cm), having sides parallel to the warp and filling directions of the cloth. The test specimen shall be arranged on a tray and placed in an oven maintained at 280 ± 2 degrees F (138 ± 1 degree C), for a period of two hours. At the end of this period, the specimen shall be removed from the oven, and conditioned under standard atmospheric conditions. The specimen shall then be measured for dimensional changes at three points of the 10 inch (25.4cm) square. The measurements shall be made at least one inch (2.5cm) from the edge of the square and at the center of the square in the warp and filling direction. The specimen shall then be visually compared with the original unheated fabric for distortion, puckering, or change in color. The change in color shall be evaluated in accordance with the evaluation specified in method 5662 of FED-STD-191. One specimen shall be tested from each sample unit. The dimensional change of the sample unit in the warp and filling direction shall be the average of the specimens tested from each direction, respectively, and shall be reported separately to the nearest 0.1 percent. Distortion and puckering shall be reported as "pass" or "fail".

4.5.2 Permanence of finish Two 20-inch (50.8cm) square specimens of the cloth shall be prepared. Using a template and indelible ink, an 18-inch (45.7cm) square shall be marked on each specimen. The specimens shall be subjected to the air permeability and thickness tests in accordance with Methods 5450 and 5030 of FED-STD-191. A container of adequate size to accommodate both specimens, prepared as described below, shall be filled to within 3 inches (7.6cm) of the top with water which shall be heated to a rapid boil. Both specimens shall be placed in the boiling water in a "skein" form prepared by stapling the two opposite sides of a specimen together to form a loop or skein. One specimen shall have the warp yarns vertical in the "skein" and the other specimen shall have the filling yarns vertical. Each specimen shall then be placed over a glass rod 1/4 inch (.6cm) in diameter and 21 inches (53.3cm) in length. A glass tube, 1/4 inch (.6cm) in diameter and 21 inches (53.3cm) in length, and approximately 100 grams in weight, shall be placed inside each loop at the bottom. Both loops shall then be suspended freely in the boiling water bath by attaching each with twine or wire to glass rods which are 1/4 inch (.6cm) in diameter and of sufficient length to rest on the top of the container. The specimen shall be subjected to the action of the boiling water bath for a period of 15 minutes, after which they shall be removed from the

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bath and allowed to drain for a few minutes, the staples shall be removed from the specimens and the specimens shall be placed flat on a horizontal screen to air dry. After the specimens are thoroughly dry they shall be exposed for at least 4 hours to a standard atmospheric condition of 65 ± 2 percent relative humidity and a temperature of 70° ± 2° F (21.7 ± 1°C). The 18-inch square (46.7cm) shall be measured to the nearest 0.01 inch (0.25mm) in 6 places, 3 in the warp direction and 3 in the filling direction. The results of the warp skein and filling skein tests for thickness, air permeability, and shrinkage shall be averaged, and acceptance based on the average results. The percentage of shrinkage in either the warp or filling direction shall be computed as follows:

$$18 - \frac{\text{distance between marks after boiling}}{18} \times 100 = \text{percent shrinkage}$$

The specimen shall again be subjected to the air permeability and thickness test to determine conformance to 3.7.1.

5. PACKAGING

5.1 Preservation-packaging Packaging shall be level A or C, as specified(see 6.2).

5.1.1 Levels A and C The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use The cloth covered by this specification is intended for use in the following:

Type I and II	- For use in fabrication of Army cargo parachutes
Type III	- For use in fabrication of special clothing (Navy) Linings in uniform clothing (Army and MC)
Type IV	- For use in linings of uniform clothing (AF, Navy, and CG)

* 6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type of cloth required (see 1.2)
c. Color of cloth required (see 3.5)
d. When first article sample is required (see 3.2)
e. When special test required for Type III cloth.
f. Width required, if other than specified (see 3.4.2)
g. Minimum length, if other than specified (see 3.6)
h. Levels of packaging and packing required (see 5.1 and 5.2)

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6.3 Standard sample For access to the standard sample, address the procuring activity issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, ILL 60684.

6.5 Changes from previous issue The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evalute the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - NU
Army - GL
AF - 11

Preparing activity:
Navy - NU

Project No. 8305-0733

Review activities:
AF - 99
DLA - DPSC-TTC
Navy - CG, MC

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